Project Topic Proposal
Adding a Role-Based Timelock Admin Module to Scaffold-ETH 2

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Target Open Source Project. The target repository is *Scaffold-ETH 2* (`https://github.com/scaffold-eth/scaffold-eth-2`). It is an open-source Ethereum development stack for building decentralized applications with smart contracts, deployment tooling, and a modern frontend. My project will contribute contract code, tests, frontend integration, and deployment support in a form that can serve as a contribution-quality extension.

Motivation and Relevance. Many Ethereum applications require privileged administrative actions, such as changing parameters, updating treasury addresses, or managing access permissions. Allowing one account to execute these actions immediately creates security risk. A safer pattern is to combine role-based permissions with a timelock so that sensitive actions must be scheduled before execution. This project is relevant because it improves security, demonstrates a realistic governance workflow, and fits naturally into an existing open-source Ethereum framework.

Proposed Contribution. I propose to add a reusable role-based timelock admin module to Scaffold-ETH 2. The module will allow authorized users to schedule sensitive actions, enforce a waiting period before execution, and support cancellation before execution. In addition to the Solidity implementation, I will integrate the workflow into the frontend, add automated tests, and provide deployment scripts and documentation for local and testnet use.

Background and Related Work. Ethereum developers often rely on common libraries such as OpenZeppelin for access control and timelock-related primitives, while Scaffold-ETH 2 provides a practical full-stack environment for decentralized application development. However, starter projects often use simplified admin patterns. This creates an opportunity to contribute a stronger reference implementation that remains understandable, deployable, and useful to developers.

Project Goals. The main goals are to implement the module, validate that it correctly enforces both permissions and delays, integrate it into the existing Scaffold-ETH 2 workflow, and demonstrate the result through local testing and public testnet deployment if feasible.

Evaluation Plan. The project will be evaluated through contract tests, integration testing, and practical demonstration. I will verify correct scheduling, cancellation, execution, permission checks, and rejection of unauthorized or premature actions. I will also validate the workflow locally and, if possible, deploy it on Sepolia. Evidence will include test outputs, screenshots, deployment logs, and a final demo or contribution-ready code package.

Planned Timeline.

Week	Planned Task
1	Study the repository, set up the environment, review relevant components, and finalize the module design.
2	Implement the contract logic, write tests, and integrate the feature into the frontend.
3	Validate locally and on Sepolia if applicable, refine the documentation, and prepare the final report and demo.

Tools and Technologies. The project will use Solidity, Scaffold-ETH 2, Foundry and/or Hardhat, TypeScript, Next.js, wagmi, viem, OpenZeppelin Contracts, Sepolia, MetaMask, and GitHub.

Conclusion. This project proposes a practical open-source contribution to Scaffold-ETH 2 by adding a reusable role-based timelock admin module. The scope is appropriate for a three-week timeline while still demonstrating technical depth in smart contract development, testing, frontend integration, and security-oriented Ethereum design.